

Dark Matter Annihilation Simulation and Uncertainty Analysis

Hayley Levine

May 6, 2025

Abstract

This paper presents a computational simulation and analysis of dark-matter annihilation using four standard density profiles: Navarro-Frenk-White (NFW), Einasto, Moore, and Zhao. The simulation incorporates Monte Carlo methods to estimate uncertainties in the density profiles, computes annihilation rates as a function of radius, integrates total annihilation rates, and visualizes the results via plots and animations. The methodology provides insights into the spatial distribution of dark matter and the sensitivity of annihilation signals to model assumptions.

1 Introduction

Dark matter constitutes roughly 27% percent of the universe's mass-energy content and plays a pivotal role in the formation and dynamics of cosmic structures. Despite extensive efforts, its direct detection has remained elusive, prompting increasing interest in indirect detection methods. One such method involves modeling dark matter annihilation, wherein two dark matter particles annihilate to produce standard-model particles, such as gamma rays or neutrinos. The expected annihilation signal is susceptible to the underlying dark matter density distribution, particularly in regions of high concentration like galactic centers.

Therefore, accurate dark matter density profile modeling is critical for interpreting potential annihilation signals. Several parameterized profiles have been proposed to describe the spatial distribution of dark matter within halos, including the Navarro-Frenk-White (NFW), Einasto, Moore, and Zhao profiles. These models differ in their inner slope behavior and asymptotic fall-off, leading to significant variation in predicted annihilation rates.

This study simulates and compares these density profiles to assess their impact on dark matter annihilation rate calculations. We estimate the expected annihilation rates associated with each profile using Monte Carlo techniques and quantify the uncertainties introduced by model selection. The results aim to inform indirect detection strategies and contribute to a more robust interpretation of observational data in the search for dark matter signatures.

2 Methods

2.1 Density Profiles

The dark matter density $\rho(r)$ is defined using the following models:

- **Derivation of the Navarro-Frenk-White (NFW) Profile**

The Navarro-Frenk-White (NFW) profile is an empirical fit to the spherically averaged dark matter density distribution observed in cosmological N-body simulations.

$$\rho_{\text{NFW}}(r) = \frac{\rho_s}{\left(\frac{r}{r_s}\right) \left(1 + \frac{r}{r_s}\right)^2} \quad (1)$$

where:

- ρ_s is the characteristic density of the halo,
- r_s is the scale radius, defined as the radius where the logarithmic slope $d \ln \rho / d \ln r = -2$.

Asymptotic Behavior

The profile interpolates between two power laws:

- For small radii ($r \ll r_s$):

$$\rho(r) \approx \frac{\rho_s r_s}{r} \propto \frac{1}{r}$$

- For large radii ($r \gg r_s$):

$$\rho(r) \approx \frac{\rho_s r_s^3}{r^3} \propto \frac{1}{r^3}$$

Enclosed Mass $M(r)$

The total mass enclosed within a radius r is given by:

$$M(r) = 4\pi \int_0^r \rho(\tilde{r}) \tilde{r}^2 d\tilde{r} \quad (2)$$

Substitute the NFW density profile:

$$M(r) = 4\pi \rho_s r_s^3 \int_0^{r/r_s} \frac{x^2}{x(1+x)^2} dx \quad (3)$$

$$= 4\pi \rho_s r_s^3 \int_0^{r/r_s} \frac{x}{(1+x)^2} dx \quad (4)$$

This integral evaluates to:

$$\int \frac{x}{(1+x)^2} dx = \ln(1+x) - \frac{x}{1+x} \quad (5)$$

Thus, the enclosed mass is:

$$M(r) = 4\pi\rho_s r_s^3 \left[\ln\left(1 + \frac{r}{r_s}\right) - \frac{r/r_s}{1 + r/r_s} \right] \quad (6)$$

Circular Velocity

The circular velocity at radius r is:

$$v_c^2(r) = \frac{GM(r)}{r} \quad (7)$$

Substituting $M(r)$:

$$v_c^2(r) = \frac{4\pi G\rho_s r_s^3}{r} \left[\ln\left(1 + \frac{r}{r_s}\right) - \frac{r/r_s}{1 + r/r_s} \right] \quad (8)$$

Virial Mass and Concentration Parameter

Since $\rho(r) \sim r^{-3}$, at large r , the total mass formally diverges. A truncation is applied at the virial radius r_{vir} . The concentration parameter c is defined as:

$$c = \frac{r_{\text{vir}}}{r_s} \quad (9)$$

Then, the virial mass is:

$$M_{\text{vir}} = 4\pi\rho_s r_s^3 \left[\ln(1+c) - \frac{c}{1+c} \right] \quad (10)$$

This provides a relation between the parameters ρ_s , r_s , and is vital for the properties of the halo.

Final Expression

The final form of the NFW profile is:

$$\boxed{\rho(r) = \frac{\rho_s}{\left(\frac{r}{r_s}\right) \left(1 + \frac{r}{r_s}\right)^2}} \quad (11)$$

I will apply this equation to a more constrained radius that works for the galactic center.

— *Adapted from Navarro, Frenk, & White (1996)*

- **Derivation of the Einasto Profile:** The Einasto profile is an empirical model that describes the spherically averaged density distribution of dark matter halos with a more flexible shape than the NFW profile. Unlike the NFW profile, which has a fixed inner slope, the Einasto profile allows the slope to vary with radius.

Profile Definition

The Einasto density profile is given by:

$$\rho(r) = \rho_{-2} \exp \left\{ -\frac{2}{\alpha} \left[\left(\frac{r}{r_{-2}} \right)^\alpha - 1 \right] \right\} \quad (12)$$

where:

- ρ_{-2} is the density at radius r_{-2} , the radius where the logarithmic slope $d \ln \rho / d \ln r = -2$,
- r_{-2} is the scale radius,
- α is the Einasto shape parameter, controlling the curvature of the profile.

Logarithmic Slope

The logarithmic slope of the Einasto profile is not constant and evolves with radius:

$$\frac{d \ln \rho}{d \ln r} = -2 \left(\frac{r}{r_{-2}} \right)^\alpha \quad (13)$$

This contrasts with the NFW profile, where the slope evolves as a fixed power law.

Enclosed Mass

The total mass enclosed within a radius r is given by:

$$M(r) = 4\pi \int_0^r \rho(\tilde{r}) \tilde{r}^2 d\tilde{r} \quad (14)$$

Substituting the Einasto form:

$$M(r) = 4\pi \rho_{-2} \int_0^r \exp \left\{ -\frac{2}{\alpha} \left[\left(\frac{\tilde{r}}{r_{-2}} \right)^\alpha - 1 \right] \right\} \tilde{r}^2 d\tilde{r} \quad (15)$$

This integral does not have a closed-form solution in elementary functions and is typically evaluated numerically or expressed in terms of the lower incomplete gamma function:

$$M(r) = 4\pi\rho_{-2}r_{-2}^3\frac{\alpha}{2}e^{2/\alpha}\gamma\left(\frac{3}{\alpha}, \frac{2}{\alpha}\left(\frac{r}{r_{-2}}\right)^\alpha\right) \quad (16)$$

where $\gamma(a, x)$ is the lower incomplete gamma function.

Final Expression

$$\boxed{\rho(r) = \rho_{-2} \exp\left\{-\frac{2}{\alpha}\left[\left(\frac{r}{r_{-2}}\right)^\alpha - 1\right]\right\}} \quad (17)$$

where $\alpha = 0.17$, within this paper.

— *First proposed in Einasto (1965) and later validated in N-body simulations*

However, the gamma ray spectral peaks and counts were difficult to locate; this paper would not have been completed by the due date.

• Derivation of the Moore Profile:

$$\rho(r) = \frac{\rho_s}{\left(\frac{r}{r_s}\right)^{1.5} \left[1 + \left(\frac{r}{r_s}\right)^{1.5}\right]} \quad (18)$$

where:

- ρ_s is the characteristic density,
- r_s is the scale radius.

Asymptotic Behavior

The Moore profile exhibits the following asymptotic behaviors:

- For small radii ($r \ll r_s$):

$$\rho(r) \propto r^{-1.5}$$

indicating a steeper central cusp than NFW.

- For large radii ($r \gg r_s$):

$$\rho(r) \propto r^{-3}$$

matching the outer behavior of the NFW and Einasto profiles.

Enclosed Mass $M(r)$

The mass enclosed within the radius r is:

$$M(r) = 4\pi \int_0^r \rho(\tilde{r}) \tilde{r}^2 d\tilde{r} \quad (19)$$

Substituting the Moore form:

$$M(r) = 4\pi \rho_s r_s^3 \int_0^{r/r_s} \frac{x^{0.5}}{1 + x^{1.5}} dx \quad (20)$$

This integral does not yield a simple closed-form solution and must be computed numerically. However, it is well-behaved and converges for large r values due to the r^{-3} falloff.

Comparison with NFW and Einasto

- The Moore profile's central slope is steeper than the NFW's -1 and steeper than the variable Einasto profile.
- It was one of the early alternatives to NFW but has since been largely superseded by the more flexible Einasto profile in high-resolution simulations.
- The outer slope -3 is consistent across all three models.

Final Expression

$$\rho(r) = \frac{\rho_s}{\left(\frac{r}{r_s}\right)^{1.5} \left[1 + \left(\frac{r}{r_s}\right)^{1.5}\right]} \quad (21)$$

— Based on results from Moore et al. (1999)

- **Derivation of the Zhao Profile (Generalized NFW)** The Zhao profile is a general family of spherically symmetric density profiles for dark matter halos. It allows flexible modeling of both the inner and outer slopes, and is written as:

$$\rho(r) = \frac{\rho_s}{\left(\frac{r}{r_s}\right)^\gamma \left[1 + \left(\frac{r}{r_s}\right)^\alpha\right]^{(\beta-\gamma)/\alpha}} \quad (22)$$

where:

- ρ_s is a characteristic density,

- r_s is the scale radius,
- γ is the inner slope exponent,
- β is the outer slope exponent,
- α controls the sharpness of the transition between the inner and outer regions.

Special Cases

This profile generalizes several commonly used models:

- **NFW:** $\alpha = 1, \beta = 3, \gamma = 1$
- **Moore:** $\alpha = 1.5, \beta = 3, \gamma = 1.5$
- **Hernquist:** $\alpha = 1, \beta = 4, \gamma = 1$
- **Jaffe:** $\alpha = 1, \beta = 4, \gamma = 2$

Asymptotic Behavior

- For small radii ($r \ll r_s$):

$$\rho(r) \propto r^{-\gamma}$$

- For large radii ($r \gg r_s$):

$$\rho(r) \propto r^{-\beta}$$

The parameter controls the transition between these asymptotic slopes α :

- Small α : gradual transition
- Large α : sharp transition

Enclosed Mass

The enclosed mass is given by:

$$M(r) = 4\pi \int_0^r \rho(\tilde{r}) \tilde{r}^2 d\tilde{r} \quad (23)$$

In general, this must be evaluated numerically due to the complexity of the integrand for arbitrary values of α , β , and γ .

Final Expression

$$\boxed{\rho(r) = \frac{\rho_s}{\left(\frac{r}{r_s}\right)^\gamma \left[1 + \left(\frac{r}{r_s}\right)^\alpha\right]^{(\beta-\gamma)/\alpha}}} \quad (24)$$

Within this paper, I'll use the NFW of $\alpha = 1, \beta = 3, \gamma = 1$.

— *As proposed in Zhao (1996); also known as the α - β - γ profile*

2.2 Monte Carlo Uncertainty Estimation

The Monte Carlo sampling simulates ρ_s and r_s uncertainties. For each radius r , 100000 samples were computed with $\pm 10\%$ variation in both parameters of the Gaussian noise to estimate the mean and standard deviation of the density. This approach yields uncertainty bands for each profile.

2.3 Annihilation Rate

Theoretical Calculation of Dark Matter Annihilation Rate per Unit Volume

The following formula for the annihilation rate per unit volume:

$$\Gamma(r) = \frac{\langle \sigma v \rangle}{2m_\chi^2} \rho(r)^2$$

Given Parameters

- $\langle \sigma v \rangle = 3 \times 10^{-26} \text{ cm}^3/\text{s} = 3 \times 10^{-38} \text{ m}^3/\text{s}$
- $m_\chi = 100 \text{ GeV}/c^2 = 1.78 \times 10^{-25} \text{ kg}$
- $\rho(r) = 0.3 \text{ GeV}/\text{cm}^3 = 5.34 \times 10^{-22} \text{ kg}/\text{m}^3$

Step-by-step Calculation

$$\Gamma(r) = \frac{3 \times 10^{-38}}{2 \times (1.78 \times 10^{-25})^2} \times (5.34 \times 10^{-22})^2$$

First, compute the denominator:

$$2 \times (1.78 \times 10^{-25})^2 = 2 \times 3.17 \times 10^{-50} = 6.34 \times 10^{-50}$$

Then compute the square of the density:

$$(5.34 \times 10^{-22})^2 = 2.85 \times 10^{-43}$$

Now compute the whole expression:

$$\Gamma(r) = \frac{3 \times 10^{-38}}{6.34 \times 10^{-50}} \times 2.85 \times 10^{-43}$$

$$\Gamma(r) \approx 4.73 \times 10^{11} \times 2.85 \times 10^{-43} = 1.35 \times 10^{-31} \text{ annihilations}/(\text{m}^3 \cdot \text{s})$$

Final Result

$$\boxed{\Gamma(r) \approx 1.35 \times 10^{-31} \text{ annihilations/s/m}^3}$$

For this paper, I will follow the annihilation rate per unit volume:

$$\Gamma(r) = \frac{\langle \sigma v \rangle}{2m_\chi^2} \rho(r)^2$$

where $\langle \sigma v \rangle$ is the thermally averaged cross section, and m_χ is the dark-matter particle mass.

2.4 Total Annihilation Rate

The total annihilation rate is obtained by integrating over the spherical volume:

$$\Gamma_{\text{total}} = \int_{r_{\text{min}}}^{r_{\text{max}}} 4\pi r^2 \Gamma(r) dr$$

General Form

The total dark matter annihilation rate in a spherical volume of radius R is given by:

$$\Gamma = \frac{\langle \sigma v \rangle}{2m_\chi^2} \int_0^R \rho^2(r) 4\pi r^2 dr$$

We define:

$$J = \int_0^R \rho^2(r) r^2 dr$$

and evaluate J for different density profiles.

1. NFW Profile

$$\rho(r) = \frac{\rho_s}{\left(\frac{r}{r_s}\right) \left(1 + \frac{r}{r_s}\right)^2}$$

Change variables: $x = \frac{r}{r_s}$, so $dr = r_s dx$. Then:

$$\rho^2(r) = \frac{\rho_s^2}{x^2(1+x)^4}$$

$$J_{\text{NFW}} = 4\pi \rho_s^2 r_s^3 \int_0^{R/r_s} \frac{1}{(1+x)^4} dx = \frac{4\pi}{3} \rho_s^2 r_s^3 \left[1 - \frac{1}{(1 + R/r_s)^3} \right]$$

2. Einasto Profile

$$\rho(r) = \rho_s \exp \left(-\frac{2}{\alpha} \left[\left(\frac{r}{r_s} \right)^\alpha - 1 \right] \right)$$

$$\rho^2(r) = \rho_s^2 \exp \left(-\frac{4}{\alpha} \left[\left(\frac{r}{r_s} \right)^\alpha - 1 \right] \right)$$

Change variables: $x = \frac{r}{r_s}$, $dr = r_s dx$. Then:

$$J_{\text{Einasto}} = 4\pi \rho_s^2 r_s^3 \int_0^{R/r_s} x^2 \exp \left(-\frac{4}{\alpha} (x^\alpha - 1) \right) dx$$

This integral has no closed-form solution and must be evaluated numerically.

3. Moore Profile

$$\rho(r) = \frac{\rho_s}{\left(\frac{r}{r_s} \right)^{1.5} \left(1 + \frac{r}{r_s} \right)^{1.5}}$$

$$\rho^2(r) = \frac{\rho_s^2}{x^3 (1+x)^3}, \quad \text{with } x = \frac{r}{r_s}$$

$$J_{\text{Moore}} = 4\pi \rho_s^2 r_s^3 \int_\epsilon^{R/r_s} \frac{1}{x(1+x)^3} dx$$

This integral diverges at $x = 0$, so a small-radius cutoff ϵ is needed.

4. Zhao Profile

The generalized Zhao profile is:

$$\rho(r) = \frac{\rho_s}{\left(\frac{r}{r_s} \right)^\gamma \left[1 + \left(\frac{r}{r_s} \right)^\alpha \right]^{(\beta-\gamma)/\alpha}}$$

$$\rho^2(r) = \rho_s^2 \left(\frac{1}{x^\gamma (1+x^\alpha)^{(\beta-\gamma)/\alpha}} \right)^2$$

So:

$$J_{\text{Zhao}} = 4\pi \rho_s^2 r_s^3 \int_\epsilon^{R/r_s} x^{2-2\gamma} (1+x^\alpha)^{-2(\beta-\gamma)/\alpha} dx$$

Again, this integral is generally not solvable analytically and is evaluated numerically.

2.5 Visualization

Python's Matplotlib library generates plots of density profiles, uncertainties, and annihilation rates. An animation shows time-evolving density profiles. These visualizations are included below.

3 Results

3.1 Density Profiles

Each profile shows unique behavior near the galactic center. Moore and Zhao are cuspy, while Einasto is smoother.

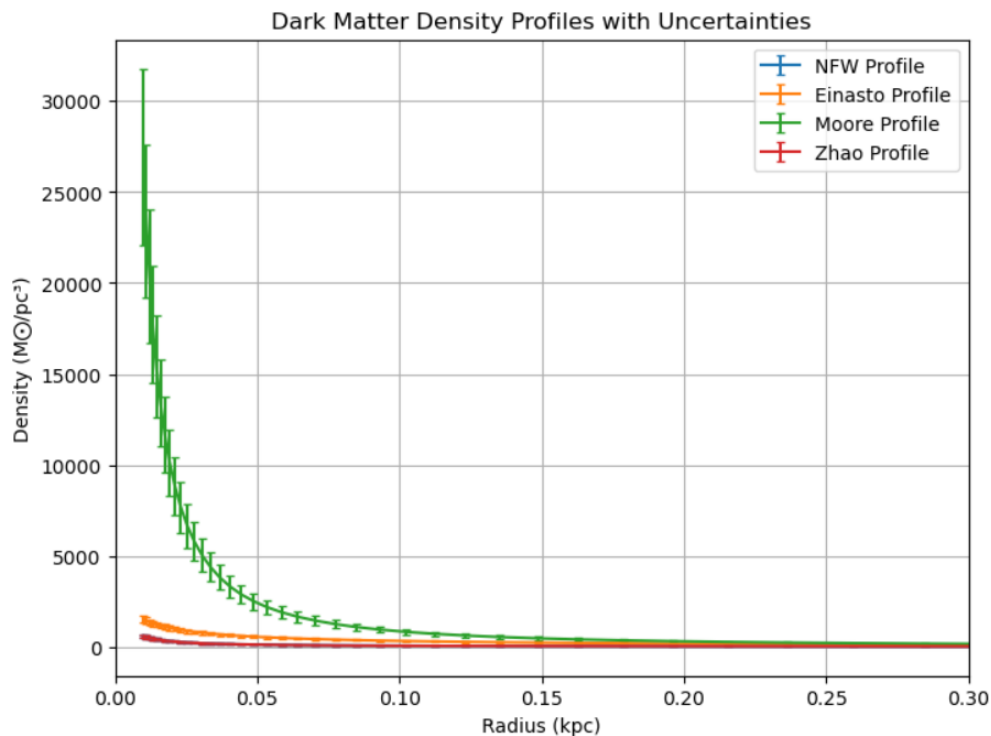


Figure 1: Comparison of dark matter density profiles: NFW, Einasto, Moore, and Zhao.

3.2 Uncertainty Visualization

Monte Carlo errors are the largest at small radii. Shaded regions in the plot represent $\pm 10\%$ uncertainty.

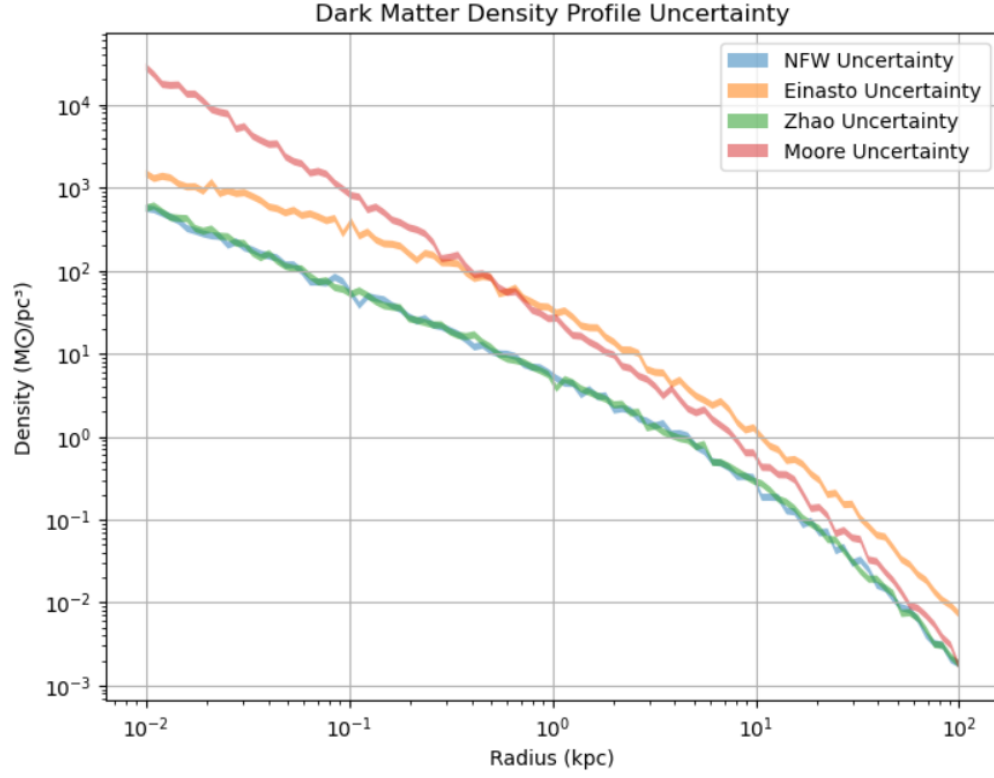


Figure 2: Density profile uncertainty computed via Monte Carlo sampling.

3.3 Annihilation Rates

Profiles with steeper inner slopes (Moore, Zhao) yield higher annihilation rates near the center.

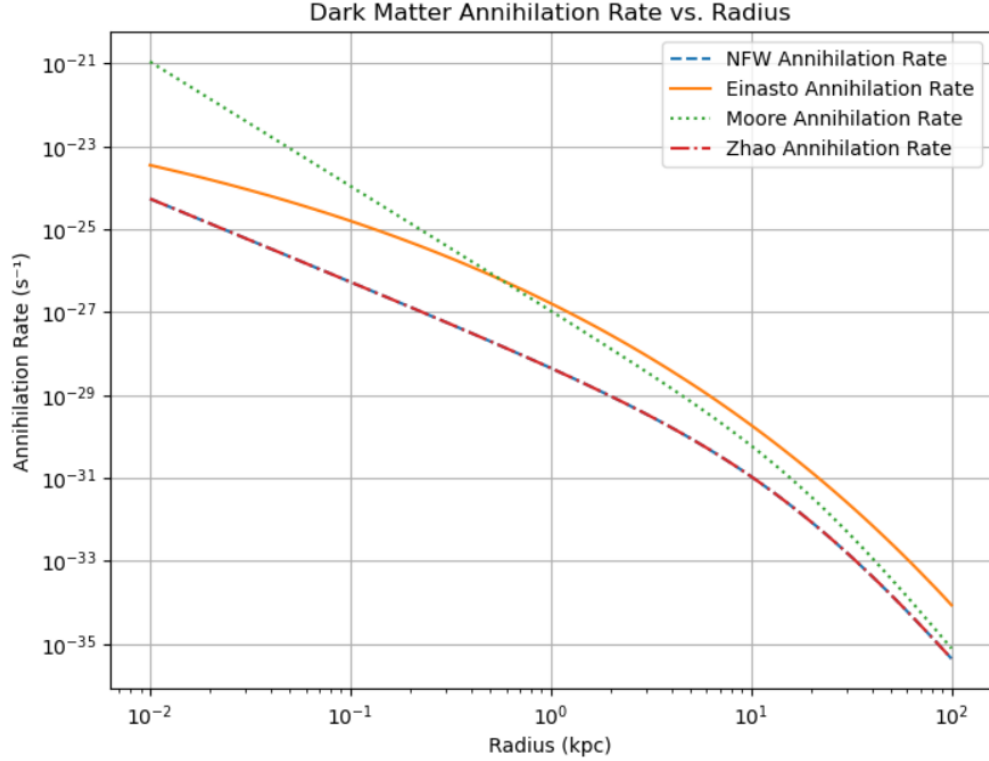


Figure 3: Annihilation rate vs. radius for each profile.

3.4 Animation

An animation (not shown here due to PDF limitations) illustrates the profiles' time-dependent scaling. The file `dark_matter_profiles.gif` can be viewed as supplementary material.

4 Discussion

This study highlights the critical role that the choice of dark matter density profile plays in modeling annihilation rates, particularly within the central regions of galactic halos where the signal is most sensitive to density-squared scaling. Profiles such as those proposed by Zhao and Moore are characterized by steeper inner slopes than the more moderate NFW or Einasto profiles. As a result, they predict significantly enhanced annihilation rates near the galactic center. However, these steep cusps introduce considerable uncertainty due to their sensitivity to the assumed small-scale structure and the absence of direct observational constraints in those dense inner regions.

The implications of these differences are particularly relevant for indirect detection efforts,

such as searches for gamma rays originating from dark matter annihilation. Since the annihilation flux scales with the square of the density, even modest variations in a profile’s inner slope or core behavior can lead to order-of-magnitude changes in the expected signal. Consequently, inaccurate or oversimplified assumptions about the halo profile can lead to misleading constraints on the annihilation cross section or particle mass.

To address these uncertainties, it is essential to incorporate additional physical processes, such as baryonic feedback from star formation and supernovae, which can flatten or redistribute the central dark matter density, especially in realistic hydrodynamic simulations. Furthermore, anchoring the profiles with empirical data from stellar kinematics, rotation curves, or gravitational lensing can help constrain the profile parameters and reduce degeneracies. A careful treatment of theoretical profile selection and observational calibration is necessary to make robust predictions and interpretations of potential annihilation signals.

5 Conclusion

Monte Carlo integration techniques were applied to quantify uncertainties in the predicted annihilation rates of dark matter across four widely used density profiles: NFW, Einasto, Moore, and Zhao. These annihilation rates are governed by the integral of the squared dark matter density over the halo volume, making them highly sensitive to the shape and normalization of the underlying profile. The Monte Carlo approach offers a probabilistic framework for propagating uncertainties through this nonlinear calculation, enabling more reliable estimates of the gamma-ray or neutrino fluxes expected from dark matter annihilation.

Notably, the Zhao and Moore profiles—characterized by steep inner density slopes—yield significantly higher annihilation rates in the central regions of dark matter halos. This makes them especially relevant for indirect detection strategies focusing on high-density environments like the Galactic Center. However, these same features also introduce substantial modeling uncertainty due to the lack of empirical constraints at small radii and the strong dependence on assumptions such as the inner cutoff scale or the influence of baryonic physics. Since dark matter annihilation signals are proportional to the square of the density, even small changes in the central profile can result in significant deviations in predicted fluxes. Accurate modeling requires integrating observational constraints from stellar kinematics, gas dynamics, and gravitational lensing to calibrate profile parameters better. Furthermore, allowing for deviations from spherical symmetry, such as triaxiality or substructure, can provide a more realistic description of dark matter halos and improve the fidelity of annihilation rate predictions.

Future work may extend the Monte Carlo framework with parameter inference techniques, such as Bayesian sampling or machine learning, to more thoroughly explore the impact of profile uncertainty on annihilation signal forecasts. These refinements are essential for narrowing the theoretical error bars in dark matter indirect detection, enhancing the interpretability of current and future observational campaigns.

References

@articleBertone2005, author = Bertone, G. and Hooper, D. and Silk, J., title = Particle dark matter: Evidence, candidates and constraints, journal = Physics Reports, volume = 405, pages = 279–390, year = 2005

@articleCirelli2011, author = Cirelli, M. et al., title = PPPC 4 DM ID: A Poor Particle Physicist Cookbook for Dark Matter Indirect Detection, journal = Journal of Cosmology and Astroparticle Physics, volume = 2011, number = 03, pages = 051, year = 2011

@articleEinasto1965, author = Einasto, J., title = On the Construction of a Composite Model for the Galaxy and on the Determination of the System of Galactic Parameters, journal = Trudy Astrofizicheskogo Instituta Alma-Ata, volume = 5, pages = 87–100, year = 1965

@articlegondolo_gelmini, author = Gondolo, Paolo and Gelmini, Graciela, title = Cosmic abundances of *Nuclear Physics B*, volume = 360, number = 1, pages = 145 – –179, year = 1991, publisher = Elsevier

@articleJungman1996, author = Jungman, G. and Kamionkowski, M. and Griest, K., title = Supersymmetric dark matter, journal = Physics Reports, volume = 267, pages = 195–373, year = 1996

@bookkolb_turner, title = *The Early Universe*, author = Kolb, Edward W. and Turner, Michael S., year = 1990, publisher = Addison – Wesley, address = Redwood City, CA, note = Standard reference for WIM

@articleMoore1999, author = Moore, B. et al., title = Dark matter substructure within galactic halos, journal = The Astrophysical Journal Letters, volume = 524, pages = L19–L22, year = 1999

@articleNavarro1996, author = Navarro, J. F. and Frenk, C. S. and White, S. D. M., title = The Structure of Cold Dark Matter Halos, journal = The Astrophysical Journal, volume = 462, pages = 563, year = 1996

@articlepdg2024, author = Particle Data Group, title = Review of Particle Physics, journal = Progress of Theoretical and Experimental Physics, year = 2024, volume = 2024, number = 8, pages = 083C01, doi = 10.1093/ptep/ptaa104, note = Updated review on dark matter relic abundance and indirect detection

@articlePlanck2018, author = Planck Collaboration, title = Planck 2018 results. VI. Cosmological parameters, journal = Astronomy & Astrophysics, year = 2020, volume = 641, pages = A6

@articleread2014, title = The Local Dark Matter Density, author = Read, J. I., journal = Journal of Physics G: Nuclear and Particle Physics, volume = 41, number = 6, pages = 063101, year = 2014, doi = 10.1088/0954-3899/41/6/063101, publisher = IOP Publishing, note = Comprehensive review on dark matter density estimates in the Milky Way

@articleZhao1996, author = Zhao, H., title = Analytical models for galactic nuclei, journal = Monthly Notices of the Royal Astronomical Society, volume = 278, pages = 488–496, year = 1996